PyGameMaker — Tutorial 9: Catch the Coins

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Catch the Coins: Win and Lose** (5 pages, about 15-20 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A complete game with a way to win and a way to lose: coins fall from the sky and you slide left and right to catch them. Catch every coin and you win. Touch the enemy and it is Game Over. From either ending, SPACE starts again.

Phase 1: A Moving Player

- [ ] **1.** Make a new project CatchTheCoins and choose the template **With Game Over Screen**. It gives you a ready-made Game Over room.
- [ ] **2.** Create a sprite spr_player (32×32) and an object obj_player with that sprite.
- [ ] **3.** Create a room room_main and place obj_player near the bottom centre.
- [ ] **4.** In obj_player: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -5*; **Right Arrow (held)** with *5*; **Keyboard: No key** with *Stop movement*.

> **You should see:** press **F5**. Left and Right move the player, and it stops when you let go. It can drift off the screen for now.

Phase 2: Coins and an Enemy

- [ ] **5.** Create sprites spr_coin (24×24, yellow circle) and spr_enemy (32×32, red square).
- [ ] **6.** Create obj_coin and obj_enemy, each with its sprite. Each gets a **Create** event: *Set vertical speed to 3*.
- [ ] **7.** Give **both** objects an **Outside Room** event with *Set variable* y to 0, so anything that falls off the bottom comes back at the top.
- [ ] **8.** In room_main, place 5 to 8 coins across the top and one enemy between them.

> **You should see:** press **F5**. Coins and the enemy fall. Touching them does nothing yet. Anything you miss comes back at the top.

Phase 3: Catching and Crashing

- [ ] **9.** In obj_player, **Create** event: *Set score to 0*.
- [ ] **10.** Add **Collision with obj_coin**: *Add to score 1*, then *Destroy other instance* (the coin, not the player!).
- [ ] **11.** Add **Collision with obj_enemy**: *Go to room* room_gameover.
- [ ] **12.** Add a **Draw** event: *Draw score at x 10, y 10*.

You should see: press **F5**. Catching a coin adds 1 and the coin disappears. Touching the enemy shows the Game Over screen, and SPACE starts again.

Phase 4: Winning the Game

- [ ] **13.** Create obj_win_text. **Draw** event: *Draw text* "YOU WIN!" at x 412, y 320 and "Press SPACE to play again" at x 340, y 400. **Key press: space**: *Restart game*.
- [ ] **14.** Create room_win with a bright background colour, and place one obj_win_text in it.
- [ ] **15.** In obj_player, add a **Step** event: *If count of* obj_coin *equals 0* then *Go to room* room_win.

You should see: press **F5**. Catch every coin and the YOU WIN! room appears. Touch the enemy and you get Game Over. SPACE starts a new game from either.

Stuck?

What is wrong	Most likely cause	What to do
The game says YOU WIN! at once	There are no coins in room_main (the count is already 0)	Place at least one coin in the room
I catch every coin but nothing happens	The Step event or the room name is wrong	Check *If count of obj_coin equals 0* and the room name room_win
A missed coin makes the game impossible to win	The coin or enemy has no **Outside Room** event	Add *Set variable* y to 0 on both
My player disappears when it catches a coin	I used *Destroy this instance*	Use **Destroy other instance**
Touching the enemy does nothing	No **Collision with obj_enemy**, or the room name is misspelled	Add it; use exactly the game-over room's name
The Game Over screen is missing	The project was not made with the "With Game Over Screen" template	Create a new project from the template
The score does not show	No **Draw** event on the player	Add *Draw score* to a Draw event
SPACE does nothing on the end screen	The end room has no object with a **Key press: space** event	Add the event and place the object in the room

Challenges

- **Try this (5 minutes):** change the falling speed, or the number of coins.
- **Push further:** add a second enemy, or make the coins and the enemy fall at different speeds.
- **Invent:** add lives, a timer, or a coin that is worth more than one point.

Vocabulary

Term	What it means
Win state	What happens when the player wins

Lose state	What happens when the player loses
Room transition	Going from one room to another (here, to the Game Over or win room)
Instance count	How many copies of an object exist right now
Template	A ready-made starting project

Check Yourself

Write your answers in the notes below.

1. How does the game know that every coin has been caught?
2. Why do missed coins need an **Outside Room** event?
3. What is the difference between the "win" room and the "Game Over" room?

My Notes